CORE JAVA SYLLABUS



Module 1: Introduction to Java

- What is Java?
- History and Features of Java
- C++ vs Java
- Java Program Structure
- First Java Program: Compilation & Execution
- JDK, JRE, and JVM
- Setting JAVA_HOME and PATH
- Java Editions and Applications
- Introduction to Java Development Tools / IDEs

Module 2: Java Basics

- Java Naming Conventions
- Comments
- Statements and Blocks
- Static and Non-static / Instance Blocks
- Identifiers and Keywords
- Literals
- Primitive Data Types and Range
- Reference / User-defined Data Types
- Variables: Primitive and Reference
- Type Casting and Default Values
- Operators

Module 3: Working with Control Structures

- Types of Control Structures
- Decision Control: if, if-else, if-else-if, switch-case
- Looping: for, while, do-while
- Enhanced for-each Loop
- Jump Statements: break, continue, return
- Nested Control Structures
- Practical: Number, series, menu-driven and pattern programs

Module 4: Input Fundamentals

- Java Program Input from Keyboard
- Scanner
- BufferedReader
- Command-Line Arguments
- Parsing Command-Line Arguments
- Basic Input Validation

Module 5: Arrays and Strings

- Single-Dimensional Arrays
- Multi-Dimensional Arrays
- Jagged Arrays
- Array Traversal and Searching
- Basic Sorting of Arrays
- String and String Literals
- String, StringBuffer and StringBuilder
- Common String Methods
- String Comparison and Manipulation

Module 6: Methods in Java

- What is a Method?
- Method Declaration and Method Signature
- Predefined and User-defined Methods
- Instance Methods
- Static Methods
- Calling Methods
- Method Overloading
- Java main() Method
- Return Type
- Parameter Passing

Module 7: Object-Oriented Programming

- Introduction to OOP
- Classes and Objects
- Instance Variables and Methods
- Constructors
- Default and Parameterized Constructors
- this Keyword
- Garbage Collection
- finalize() concept and modern Java considerations

Module 8: Inheritance and Polymorphism

- Introduction to Inheritance
- Single, Multilevel and Hierarchical Inheritance
- Hybrid Inheritance through supported Java mechanisms
- super Keyword
- Method Overriding
- final Keyword: Variable, Method and Class

- Dynamic Method Dispatch
- Compile-time and Runtime Polymorphism
- Practical: Inheritance and polymorphism programs

Module 9: Abstraction and Encapsulation

- Abstraction
- Abstract Methods
- Abstract Classes
- Interfaces
- Interface vs Abstract Class
- Interface Implementation
- Multiple Interfaces
- Encapsulation
- Getters and Setters
- Access Modifiers
- Packages and import Statement

Module 10: Exception Handling

- Introduction to Exceptions
- Checked vs Unchecked Exceptions
- try, catch and finally
- Multiple catch Blocks
- Nested try
- throw and throws
- Custom Exceptions
- Chained Exceptions
- Exception Propagation
- Practical: Robust programs with exception handling

Module 11: Java Input/Output and File Handling

- Java I/O Streams
- Byte Streams vs Character Streams
- Buffered Streams
- File Class
- Text File vs Binary File
- Reading and Writing Files
- InputStreamReader and BufferedReader
- Data Streams
- Serialization and Deserialization
- Stream Class Hierarchy
- Practical: File copy, read/write and serialization programs

Module 12: Collections Framework

- Wrapper Classes
- Autoboxing and Unboxing
- Collection Framework and Hierarchy
- List, Set and Map
- ArrayList
- LinkedList
- HashSet
- TreeSet
- HashMap
- TreeMap
- LinkedHashMap
- Iterators
- Enhanced for Loop
- Generics
- Comparable and Comparator

Module 13: Multithreading

- Introduction to Threads
- Thread Lifecycle
- Creating a Thread by Extending Thread
- Runnable Interface
- run() and start() Methods
- Thread Class Constructors and Methods
- Thread Priorities and Sleep
- Synchronization Method and Block
- Inter-thread Communication

Suggested Mini Projects

- Student Management System
- Banking & ATM System
- Employee Payroll System
- Library Management System
- Inventory Management System
- Course & Training Management System
- Hospital Management System
- E-Commerce Shopping System
- Bus/Railway Reservation System
- Online Examination System